

Unit 11 Test Study Guide

(Radical Expressions & Equations)

Name: _____

Date: _____ Per: _____

Topic 1: Simplifying Radicals

Perfect Squares: _____

Perfect Cubes: _____

1. $5\sqrt{112}$

2. $-\sqrt{44a^7b^{16}}$

3. $2\sqrt{108x^9y^2z^{14}}$

4. $\sqrt[3]{189}$

5. $\sqrt[3]{-16a^4}$

6. $\sqrt[3]{320m^{15}n^7}$

Topic 2: Adding & Subtracting Radicals

7. $5\sqrt{7} + 2\sqrt{7}$

8. $\sqrt{72} - \sqrt{18}$

9. $4\sqrt{48} - \sqrt{147} - 3\sqrt{50}$

10. $3\sqrt{5} + 7\sqrt{125} - 2\sqrt{45}$

Topic 3: Multiplying Radicals

11. $5\sqrt{6} \cdot \sqrt{27}$

12. $7\sqrt{10} \cdot -3\sqrt{18}$

13. $\sqrt{5x^2} \cdot \sqrt{8x^3}$

14. $\sqrt{6}(7\sqrt{3}+6)$

15. $3\sqrt{2}(5\sqrt{10}-2\sqrt{6})$

16. $(5+2\sqrt{7})(4-\sqrt{7})$

17. $(1+2\sqrt{8})^2$

Topic 4: Dividing Radicals

18. $\sqrt{\frac{48}{3}}$

19. $\frac{\sqrt{18}}{\sqrt{75}}$

20. $\frac{6\sqrt{56}}{2\sqrt{63}}$

21. $\sqrt{\frac{12}{5}}$

22. $\frac{2}{6\sqrt{2}}$

23. $\frac{20\sqrt{8}}{4\sqrt{3}}$

24. $\frac{5-\sqrt{3}}{\sqrt{12}}$

25. $\frac{\sqrt{8}}{4+\sqrt{6}}$

Topic 5: Solving Radical Equations

26. $-2\sqrt{x} = -16$

27. $\sqrt{4x+1} + 6 = 11$

28. $x = \sqrt{56-x}$

29. $p-3 = \sqrt{5p-15}$